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AI for Bharat Hackathon

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Team Name : Team Spouse

Team Leader Name : Sreyashi Deychaki

Problem Statement : AI for Retail, Commerce & Market Intelligence - Transforming Retail Category Management with Autonomous Agents

Brief about the Idea:

Category Manager
Owner of category
GMV, Margin & CX

What they manage

Right Assortment | Right Price | Right Inventory | Customer Experience

→ Millions of shoppers

Why it's hard
(Bharat context)

- Fragmented regional demand
- High price sensitivity
- Constant operational constraints

Reality today
45–60% of time lost
(~50–55 hrs/week)

- Data pulls
- Reports & decks
- Explaining the past

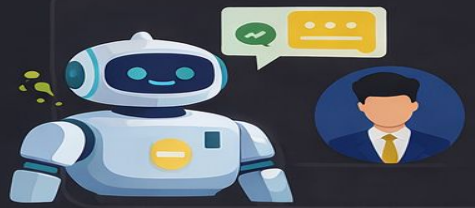
Core Insight
Category Managers are
data-rich, but insight-poor

Impact

- Reactive decisions
- Missed revenue & margin
- Late CX issue detection

The Solution: AI-CM

An Agentic Decision Intelligence Platform



AI-CM is a ²Multi-Agent System²
that acts as an autonomous team for the Category Manager.

- **From Dashboard to Copilot:** It doesn't just display charts; it actively **'investigates'** anomalies.

Core Workflow: Ask → Query → Reason → Act

Ask: "Why did margin drop?"

Query: (Agent writes SQL)

Reason: (Agent correlates with Competitor Price)

Reason: (Agent correlates Competitor Price)

Act: (Agent proposes **Price Match**)

Core Shift: From "Reviewing Reports" (Reactive) to "Approving Actions"

Reviewing Reports
(Reactive)

Approving Actions
(Proactive)

How is it different from existing solutions?

What Exists Today

Platforms like **Blue Yonder, Symphony RetailAI, Tableau, Power BI**

- Demand forecasting
- Replenishment & inventory planning
- SKU performance analytics
- Location-based reporting
- Historical dashboards

Strong in their own domain, but operate as **discrete systems**

The Gap

- **Category** Managers **do NOT** manage problems **in isolation**
- They manage the **entire lifecycle of a category & its SKUs**
- Insights are fragmented across tools and dashboards

No single solution provides an **end-to-end, category-first view** designed around the daily workflow of a Category Manager.

How AI-CM is Different

- **AI-CM is not another analytics or forecasting tool.**
- It is a **Category Manager Copilot** that **unifies all category decisions** into one continuous lifecycle view.

One login. One system. One source of truth for the category.

How will AI-CM solve the problem?

- **AI-CM will:**
  - Consolidate **insights across** assortment, pricing, inventory, demand, and customer feedback
  - **Explain & recommend** clear **next actions** at the SKU and category level
  - **Accelerate category decisions** to deliver **faster cycle time** and higher-quality outcomes

Large Company	Mid-Size Company	Small Company
Current: 40–50 hrs/week ➡ 20–25 hrs/week	Current: 25–30 hrs/week ➡ 12–15 hrs/week	Current: 12–15 hrs/week ➡ 5–7 hrs/week
High-Impact outcomes	**High-Impact outcomes**	**High-Impact outcomes**
Predictive planning	Predictive planning	Predictive planning
Revenue & margin gains	Revenue & margin gains	Revenue & margin gains
Consistent CX	Consistent CX	Consistent CX

50–60% Cycle Time Savings for Category Managers, Plus Predictive, **Business-Impact Decisions**

USP of AI-CM

The end-to-end platform for data-driven insights and action-oriented workflows

Autonomous Investigation (The Analyst Agent)

CM Asks: "Show me top 5 low-stock SKUs."

AI Action: Instantly writes SQL, queries DB, and renders a table. No Analyst needed.

Low-Stock SKUs

ID	SKU Name	Current Stock	Last Week Avg. Sales	Days of Sp
1	Pressure Cooker Elite 1L	129	49	5
2	Laptop Desk with Fan	175	45	5
3	Bluetooth Headphones X3	179	12	4
4	Compact Mixer Grinder X2	105	23	5
5	Air Purifier Pro Black	104	36	5

Deep Reasoning (The Strategist Agent)

System Adds: "Sales dropped because Competitor X cut prices by 15% *yesterday*"

Tech: Chain-of-Thought reasoning + RAG (*Business Context*)

Sales Alert!

Sales dropped because **Competitor** X cut prices by 15% yesterday.

Prescriptive Actions (The Planner Agent)

Proposal: "I recommend a **10% price drop** to regain Volume Share.

Est. Margin Impact: -2%.

Sales Strategy Proposal Approve

Sales Strategy = 42%

Est. Margin Impact: -2%

Seller Communication

The Liaison Agent)

Action: Automatically drafts and serids compliance emails to Sellers with poor metrics.

AI-CM

– Copilot for Category Managers

Large Company Outlook | Chat | PDFs & Excel → External | PDF & Excel

Answer-Refine-Act Copilot = Enhanced Productivity for Category Managers

Key Features of AI-CM –

scalable phased rollout for category managers

MVP (Foundation):

- **Goal:** Read-Only Intelligence
- **Scope:**
  - **Ingestion Pipeline**
  - **Analyst Agent** (Text-to-SQL)
  - Strategist Agent (Why?),
  - Chat UI

Phase 1 (Action):

- **Goal:** Closed-Loop Execution
- **Scope:**
  - Planner Agent (Approvals)
  - Liaison Agent (Emails)
  - Forecasting Service (Python)
  - Watchdog Agent

Phase 2 (Expansion):

- **Goal: Customization**
- **Scope:**
  - Agent Studio (No-Code Builder)
  - Feedback Learning Loop

Phased Value Quickly → Scale Capabilities Over Time

Why Agentic AI?

Why not just a Chatbot wrapper?

1. **Tool Use & Grounding:**
   - LLMs hallucinate. **Agents** execute tools (SQL, Search, API): If the tool returns an error, they self-correct.
2. **Multi-Step Reasoning:**
   - Complex problems (**"Diagnose Margin Drop"**) require a chain: Get Data → Check Competitors → **Check Inventory**. Agents persist state across this chain.
3. **Active Execution:**
   - Chatbots just talk. **Agents can do** (Send Emails, Update ERPs).
4. **Self-Correction:**
   - ReAct pattern allows the Agent to fix its own bad SQL queries without user intervention.

Business Value

- **For the Business:**
  - **Revenue:** 5-10% uplifts by capturing stock-out/pricing opportunities instantly.
  - **Margin:** Protection through smarter Markdown management.
- **For the User (Category Manager):**
  - **Speed:** 3-5 Day decision lag → Near Real-time.
  - **Efficiency:** Automated Data Extraction (saves 20hrs/week).
  - **Focus:** Shift from "Data Stitching" to "Strategy".

What are the use cases of the AI-CM Platform?

AI-CM

Ask your question...

Welcome, CM Prakash CP

- Dashboard
- News
- Approvals 3
- Reports
- Alerts
- Configuration
- Support

SKU Performance Focus | Category: Baby Care | Subcategory: All | Next 7 Days | Export

SKU Performance Focus

Revenue | Pricing | Skus | Drops | Insights

Next 7-Day Prediction $1.32M +1.5%

Alerts & Insights

Cradles Sales Alert
Competitor price drop detected. Recommend matching price.
View Policy

Inventory Alert
Stockout risk for SKU-55 in South Region.
View Details

AI-CM Copilot

Hi CM Prakash,
What can I assist you with today?

Insight:
Sales growth in the South region is mainly driven by:
- Cradles (+27.5%)
- Diapers (+11.1%)
- Toys (+5.6%)

Warning:
Product sales of Cradles are low in East India.
What's the primary reason?

There are more family caregivers in East India, while in South India, babies are mostly growing with caregivers hence they want kids and toys.

What are next best steps?
Send

Top SKU Movers

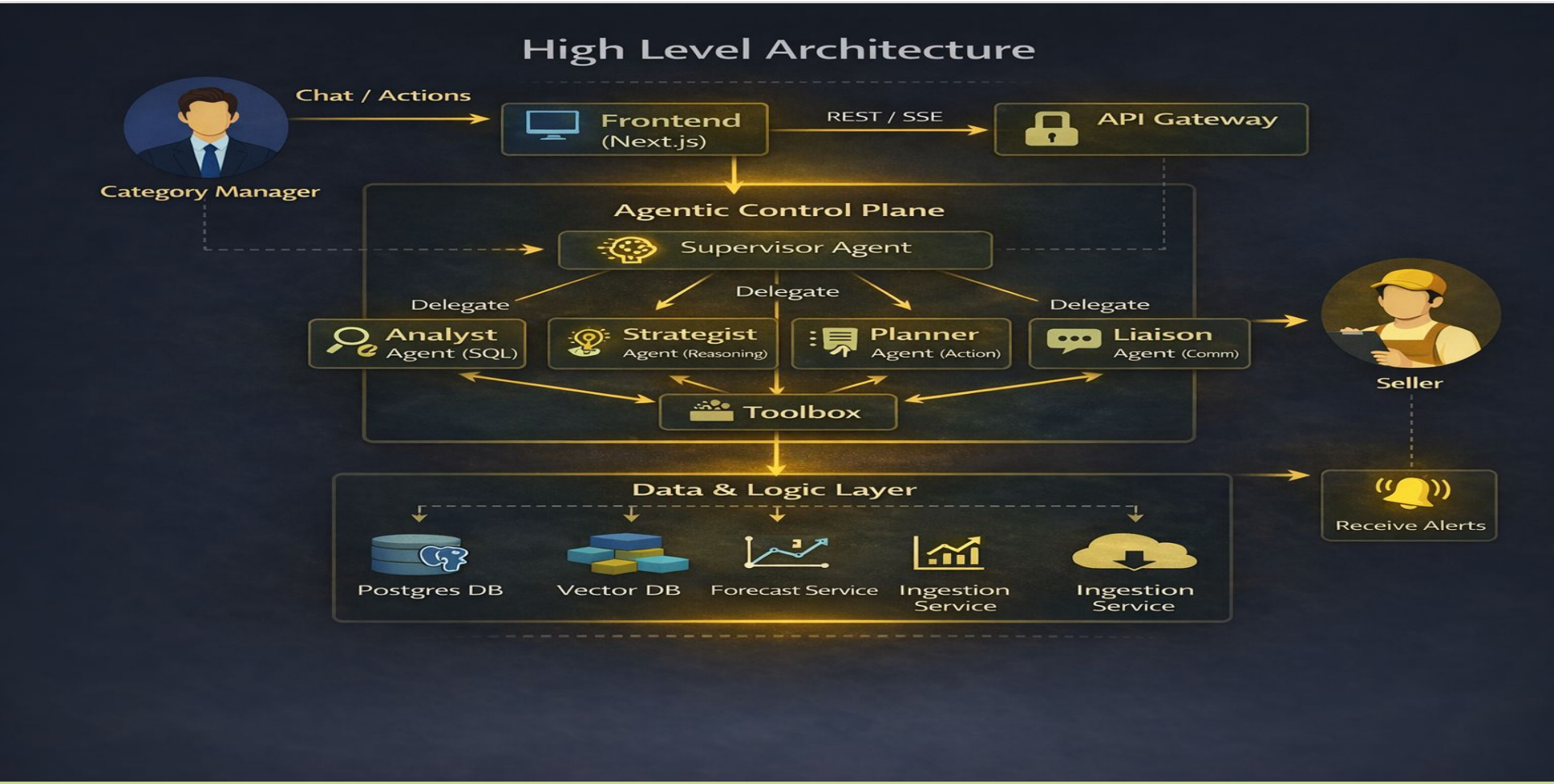
More >

$345.3k +$18.1k

Cradles	$345.3k	South
Toy Cars	$102.5k	South
Feeding Bottles	$108.4k	South

Performance by Region

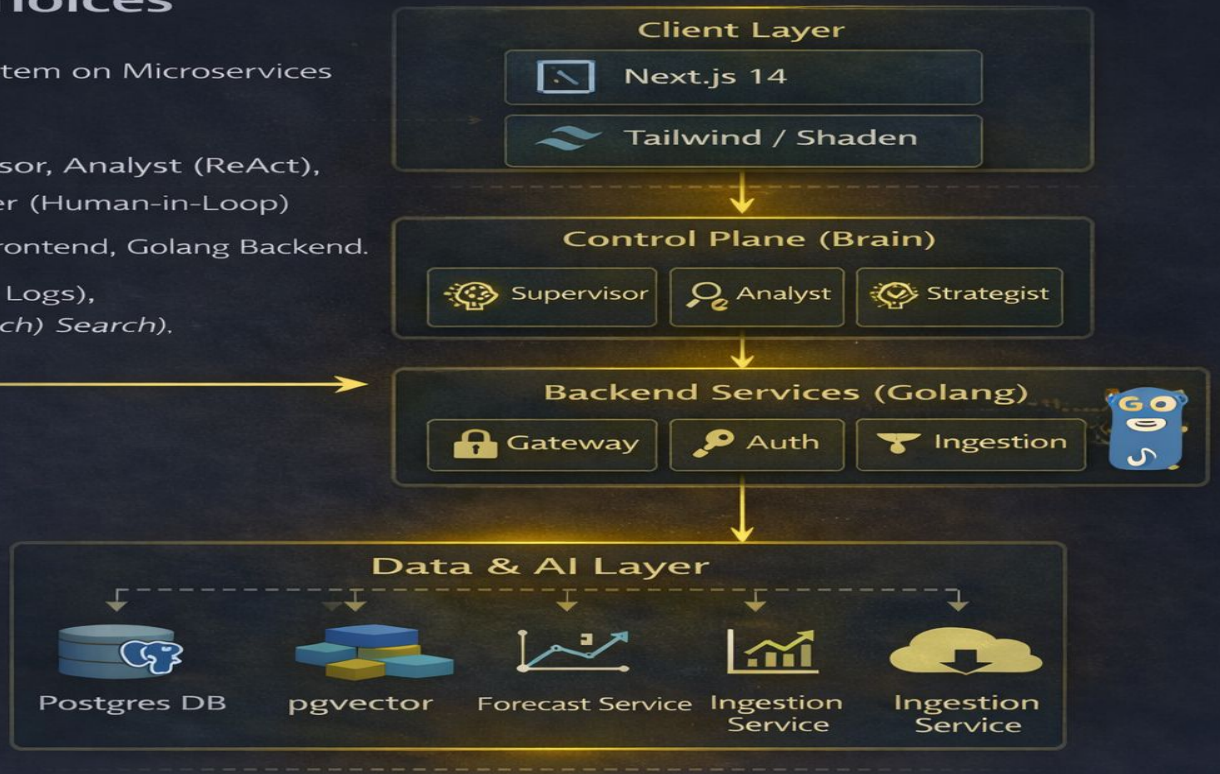
South Region
$331.4k


High Level Architecture
Chat / Actions
Category Manager
Frontend (Next.js)
REST / SSE
API Gateway
Agentic Control Plane
Supervisor Agent
Delegate
Delegate
Delegate
Analyst Agent (SQL)
Strategist Agent (Reasoning)
Planner Agent (Action)
Liaison Agent (Comm)
Toolbox
Seller
Receive Alerts
Data & Logic Layer
Postgres DB
Vector DB
Forecast Service
Ingestion Service
Ingestion Service

Technology Choices

- **Pattern:** Multi-Agent System on Microservices
- **Core Components:**
  - **Control Plane:** Supervisor, Analyst (ReAct), Strategist (RAG), Planner (Human-in-Loop)
  - **Microservices:** Next.js Frontend, Golang Backend.
  - **Data Layer:** MinIO (Raw Logs), +fybitor Search) Search).

Tech Stack:



Cost Estimation

Core Infrastructure Costs

Resource	Service	Details	Est. Cost/Mo
Compute	AWS Fargate	Control Plane + API (Optimized)	~$50
Database	RDS PostgreSQL	db.tag.small (Posgres + priescer)	~$60
Storage	Amazon S3	Raw Logs & Backups	~$5
Networking	ALB / NAT	Load Balancer & Data Transfer	~$15
Total Infra	Core Platform Running Cost		~$130

Cost Breakdown (Per Month Buffer)

Item	Volume	Unit Price	Est. Cost
Input Tokens	5M Tokens (Context)	$3 / 1M	~$15
Output Tokens	1M Tokens (Reasoning)	$15 / 1M	~$15
Total AI			~$30 ~$50

Cost Scaling Scenarios

Stage	Daily Queries	Implied Infra	Total Est.
Startup	~50	$130	~$160
Growth(5x)	~250	$180	~$330
Scale (20x)	~1000	~$400	~$1000

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Thank You